

Smart Home Energy Advisor

Technical Reference Documentation

Project Overview

Data pipeline internals, three-dataset EDA, feature augmentation, model comparison, Egyptian tariff and budget engine, recommendations, what-if analysis, and Streamlit architecture.

EDA

Preprocessing

ML Modeling

Tariff & Budget

Deployment

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1. Project Objective & Scope

The Smart Home Energy Advisor estimates appliance electricity consumption, converts the estimate into an Egyptian electricity bill, compares it with a user-declared monthly budget, reports cost per resident, and generates reduction advice plus a what-if saving scenario.

Learning task: supervised regression; target = Energy Consumption (kWh) per Dataset 1 appliance reading.

Current monthly conversion: `run_smart_home_advisor()` treats the model output as hourly kWh and multiplies it by 24×30 to estimate 30-day consumption.

Current modeling sources: Dataset 1 is the labeled source; Dataset 3 supplies auxiliary hardware metadata; Dataset 2 remains supporting EDA.

Non-goals: the current notebook does not implement time-series forecasting or real smart-meter/device ingestion.

2. Data Sources & Provenance

Dataset	Rows × Columns	Role
Dataset 1 — Smart Home Energy Consumption	100,000 × 8	Only labeled training source; target + 10 appliance types.
Dataset 2 — Smart Home Energy Usage	1,000,000 × 10	Behavioural/telemetry-style EDA: occupancy, duration, temperature setting, timestamp
Dataset 3 — Household Appliances Metadata	33 × 10 raw	Power_max, available_duration, standardized appliance and category metadata.

Dataset 1: Home ID, Appliance Type, Energy Consumption, Time, Date, Outdoor Temperature, Season, Household Size.

Dataset 2: timestamp, home_id, energy_consumption_kWh, temperature_setting_C, occupancy_status, appliance, usage_duration_minutes, season, day_of_week, holiday.

Dataset 3: id, first_ts, last_ts, available_duration, plug_name, standardized_appliance_name, appliance_category, comment, files_names, power_max.

Seven Dataset 1 appliance names map to Dataset 3: Fridge, Dishwasher, Microwave, Air Conditioning, Computer, TV and Washing Machine.

3. Data Quality Assessment

Check	Dataset 1	Dataset 2	Dataset 3
Missing values	0	0	13 in comment
Exact duplicates	0	0	0
Per-appliance target outliers	0	0	—
Metadata IQR scan	—	—	power_max 0; available_duration 4
Cleaned shape	100,000 × 8	1,000,000 × 10	33 × 9

Dataset 3 drops comment because it is the only column with missing values. The four available_duration IQR flags are retained as legitimate short monitoring windows.

Dataset 1 has 7,997 potential global IQR outliers, but the notebook evaluates the target within appliance groups; the per-appliance scan reports 0. This avoids treating a naturally high-draw appliance as an outlier relative to a low-draw appliance.

4. Exploratory Data Analysis

4.1 Dataset 1 — Consumption Behaviour

Mean consumption is ≈ 1.500 kWh (SD ≈ 1.181), ranging from 0.10 to 5.00 kWh.

Appliance	Mean kWh
Air Conditioning	3.4999
Heater	3.4868
Dishwasher	1.1034
Oven	1.1031
Washing Machine	1.1014
Microwave	1.1010
TV	1.0972
Computer	1.0955
Lights	1.0874
Fridge	0.2983

Spring has the highest seasonal mean (≈ 1.5109 kWh). Household size 2 is highest (≈ 1.5132 kWh), but the relationship is non-monotonic. Outdoor temperature correlation is ≈ 0.0020 . Hour 14 is highest at ≈ 1.5388 kWh.

The new home-level analysis finds 500 homes; the highest-average home in the printed top-10 table is Home 193 at ≈ 1.7549 kWh across 187 readings.

The hour $\times$ appliance analysis shows that appliance identity strongly separates consumption bands while within-appliance hourly changes are relatively small.

4.2 Dataset 2 — Usage Logs

Aspect	Result
Mean energy	≈2.5488 kWh
Highest mean appliance	Dishwasher ≈2.5552 kWh
Occupied mean	≈2.5498 kWh
Unoccupied mean	≈2.5479 kWh
Highest mean season	Winter ≈2.5524 kWh
Temperature-setting correlation	≈−0.000254
Duration/energy correlation	≈0.000486
Highest mean hour	Hour 5 ≈2.5601 kWh
Holiday mean	≈2.5529 kWh vs ≈2.5484 non-holiday

Average usage duration is ≈59.5 minutes. The new timestamp analysis shows only small month/weekday/weekend differences; Friday has the highest weekday mean at ≈2.5582 kWh.

The notebook's timestamp range is 2023-01-01 to 2137-01-29. This is a characteristic of the supplied data and should not be silently reinterpreted as a normal one-year period.

The appliance × occupancy analysis also shows small differences. For example, Dishwasher is ≈2.5568 kWh occupied vs ≈2.5537 unoccupied.

4.3 Dataset 3 — Appliance Metadata

Metric	Result
Standardized appliance names	17
Categories	4
Mean power	≈1,246.08 W
Power range	16–3,267 W
Highest mean power	Dryer ≈3,267 W
Mean availability	≈29.41 days
Highest availability	Air conditioner = 30 days
Lowest availability	Screen ≈21.37 days
Most represented appliance	Washing machine = 6 records

The updated notebook derives $\text{theoretical_energy_Wh} = \text{power_max} \times \text{available_duration} \times 24$. This is a theoretical upper-bound indicator, not measured consumption. Dryer, boiler and washing machine have the largest values under this calculation.

4.4 Consolidated EDA Insights

Across Dataset 1, appliance identity dominates the available linear signal. Temperature, household size, hour and month are weak. Dataset 2 likewise shows near-zero linear correlations for temperature setting and duration.

The deeper EDA strengthens diagnostic understanding but does not automatically make Dataset 2 a model input. The final estimator still uses Dataset 1 plus selected Dataset 3 metadata.

5. Preprocessing Pipeline

Dataset 1

- Missing values: none.
- Duplicates: 0 removed.
- Per-appliance IQR scan: 0 outliers.
- Hour, Month, DayOfWeek and IsWeekend engineered.
- Time and Date dropped.
- Appliance Type, Season and DayOfWeek one-hot encoded with drop_first=True.
- Home ID excluded from X.

Train/test split: 80,000 / 20,000 rows, random_state=42; 23 features. StandardScaler is fitted only on training data for Outdoor Temperature, Household Size, Hour and Month.

Dataset 2

- Missing values: none.
- Duplicates: 0.
- Per-appliance IQR scan: 0.
- Hour, Month and IsWeekend derived.
- Occupancy mapped to 1/0.
- Appliance, season and day_of_week one-hot encoded.
- home_id excluded.
- 800,000 / 200,000 train/test rows; 21 features.

Scaled Dataset 2 columns: temperature_setting_C, usage_duration_minutes, Hour and Month.

Dataset 3

comment dropped; id/first_ts/last_ts/plug_name/files_names dropped; standardized appliance and category one-hot encoded; power_max and available_duration standardized.

Leakage prevention is explicitly checked: training scaled means are approximately zero, while test means are not exactly zero, confirming the test split was transformed rather than used to fit the scaler.

6. Feature Augmentation from Dataset 3

The notebook joins hardware metadata to Dataset 1 using seven appliance mappings. It adds three features: `power_max`, `available_duration` and `Auxiliary_Data_Available`, increasing the feature count from 23 to 26.

Appliance	power_max W	days	Aux
Washing Machine	2430.67	29.74	1
Dishwasher	1806	30	1
Microwave	1757	30	1
Fridge	864.75	30	1
Computer	280	30	1
TV	54	30	1
Air Conditioning	45	30	1
Oven / Heater / Lights	864.75 median	30 median	0

Unmatched appliances receive median fills: 864.75 W and 30 days, while `Auxiliary_Data_Available=0` tells the model the metadata was not directly matched.

The notebook notes two auxiliary-source weaknesses: Fridge's value coincides with the median fill, and Air Conditioning's 45 W appears to be a low-draw plug-level value rather than a compressor nameplate rating.

7. Modeling Methodology

The task is regression on per-reading kWh. Metrics are MAE, RMSE and R^2 ; R^2 is used for programmatic model selection. Baseline uses 23 Dataset 1 features; augmented uses 26 features after Dataset 3 metadata is added.

Model	Configuration
Linear Regression	scikit-learn defaults
Random Forest	50 trees; max_depth=10; n_jobs=-1; random_state=42
XGBoost	50 trees; depth=5; learning_rate=0.1; squared-error objective; n_jobs=-1
Gradient Boosting	50 estimators; depth=3; learning_rate=0.1; random_state=42

The notebook uses a shared `evaluate_models()` helper so every candidate is trained and evaluated through the same path. The best name is selected with `results_df["R2"].idxmax()`.

8. Full Results — Baseline vs Augmented

Model	MAE	RMSE	Augmented R ²
Linear Regression	0.475984	0.583974	0.757542
Gradient Boosting	0.476046	0.584093	0.757443
Random Forest	0.476267	0.584930	0.756748
XGBoost	0.476415	0.585176	0.756543

Model	Baseline R ²	Augmented R ²	Change
Linear Regression	0.757542	0.757542	0.000000
Random Forest	0.756897	0.756748	−0.000149
XGBoost	0.646295	0.756543	+0.110248
Gradient Boosting	0.537733	0.757443	+0.219710

Dataset 3 augmentation substantially lifts XGBoost and Gradient Boosting, while Linear Regression remains unchanged and Random Forest changes only marginally. After augmentation, all four models converge around $R^2 \approx 0.757$.

9. Best Model & Feature Influence

The notebook selects Linear Regression on the 26-feature augmented set: $R^2 = 0.7575$, MAE = 0.476 kWh, RMSE = 0.584 kWh on 20,000 held-out rows.

The actual-vs-predicted plot shows appliance-specific bands; residuals are centered near zero without an obvious funnel pattern. RMSE/MAE ≈ 1.23 is a descriptive indicator that errors are not dominated by a small number of extreme misses.

Feature	Coefficient
Appliance Type_TV	-2.3859
Appliance Type_Computer	-2.0327
Appliance Type_Fridge	-1.9167
Appliance Type_Heater	+1.5117
power_max	-1.1772
Appliance Type_Lights	-0.8829
Appliance Type_Oven	-0.8726
available_duration	-0.3676
Appliance Type_Dishwasher	+0.3591
Appliance Type_Microwave	+0.2850
Auxiliary_Data_Available	+0.2437
Appliance Type_Washing Machine	+0.1104
Season_Winter	-0.0415
Season_Spring	-0.0219
Season_Summer	-0.0146

Interpretation warning: the negative power_max coefficient must not be interpreted as higher-rated appliances consuming less. power_max is tied to appliance identity, so its coefficient can act as a collinearity correction.

10. Egyptian Tariff & Budget

Monthly kWh	Notebook calculation
0–50	$0.68 \times \text{total}$
51–100	$50 \times 0.68 + \text{remainder} \times 0.78$
101–200	$0.95 \times \text{total}$
201–350	$200 \times 0.95 + \text{remainder} \times 1.55$
351–650	$200 \times 0.95 + 150 \times 1.55 + \text{remainder} \times 1.95$
651–1000	$2.10 \times \text{total}$
>1000	$2.23 \times \text{total}$

The tariff is hard-coded in `calculate_egyptian_tariff()`. In the updated master pipeline it is applied to the **monthly** kWh estimate rather than directly to one appliance-reading prediction.

Budget status: **Within Budget** below 90%, **Close to Budget** from 90% through 110%, and **Over Budget** above 110%. Recommendations trigger at 90% or higher.

The updated evaluator accepts `num_persons` and exposes `bill_per_person` as `cost_per_person` in the final payload.

11. Recommendations, What-if & Master Pipeline

`generate_recommendations()` receives a `top_appliance` argument. If it is missing, the current function defaults to Air Conditioner. At $\geq 90\%$ of budget it recommends reducing operating hours, using energy-saving settings, and avoiding standby draw.

The what-if scenario reduces predicted monthly kWh by 15%, then recalculates the tariff and reports the resulting bill and EGP saving. The bill is therefore not assumed to fall by exactly 15%.

`run_smart_home_advisor()` performs: model prediction $\rightarrow$ hourly-to-monthly scaling $\rightarrow$ tariff calculation $\rightarrow$ budget evaluation $\rightarrow$ recommendations $\rightarrow$ what-if scenario $\rightarrow$ payload.

Payload	Meaning
<code>best_model_name / accuracy_r2</code>	Current model provenance; Linear Regression / ≈ 0.7575
<code>number_of_residents</code>	Resident count used for cost-per-person calculation
<code>predicted_kwh</code>	Hourly prediction $\times 24 \times 30$
<code>estimated_bill</code>	Monthly tariff result in EGP
<code>cost_per_person</code>	Estimated bill / residents
<code>budget_status / message</code>	Budget classification and explanation
<code>recommendations</code>	Generated actions
<code>what_if_scenario</code>	15% reduction scenario with recomputed bill/saving

Example: the notebook's sample prediction is 1.11 kWh. The master pipeline scales this to about 799.2 kWh/month. This illustrates the current assumption, not a measured household total.

12. Deployment Architecture

Component	Responsibility
train_model.py	Offline training and artifact generation.
advisor_logic.py	Loads artifacts and contains prediction, tariff, budget and recommendation logic.
app.py	Streamlit presentation layer; no model training at runtime.
Artifacts	Serialized model/scalers/metadata used for reproducible inference.

The application surface can expose appliance selection, usage level, household/resident information and monthly budget, then render estimated kWh, bill, budget status, cost per person, recommendations and the what-if scenario.

The current model still does not consume Dataset 2 telemetry fields. Those variables are available for future feature integration but should not be documented as active model inputs until they are actually merged and retrained.